B3.1 Gas exchange

NAME

DATE

How to use this sheet: go through the slides and videos with it, then close them and write from memory. Use your own words and add page numbers from the book where you want more detail.

Guiding questions. What adaptations let multicellular organisms exchange gases? How do flowering plants and mammals differ, and resemble each other, in gas exchange? (The plant half is in Plant Systems.)

B3.1.1 Why every organism needs gas exchange

TEXTBOOK PP. 273–274

Why do cells need a constant supply of oxygen and removal of carbon dioxide?

Give two reasons why a large animal cannot rely on diffusion across its body surface.

B3.1.2 What makes a good gas-exchange surface

TEXTBOOK P. 274

PROPERTY	HOW IT HELPS GASES MOVE
Permeable	
Thin	
Moist	
Large surface area	

B3.1.3 Keeping concentration gradients going

TEXTBOOK P. 275

Why would diffusion stop if the gradient were not maintained?

FEATURE	HOW IT MAINTAINS THE GRADIENT
Dense network of blood vessels	
Continuous blood flow	
Ventilation with air (lungs)	
Ventilation with water (gills)	

B3.1.4 How mammalian lungs are adapted

TEXTBOOK PP. 276–279

Draw two or three alveoli with their capillaries. Annotate: **one-cell-thick wall**, **moist lining**, **surfactant**, **capillary bed**, **bronchiole**, direction of **O_2** and **CO_2**.

ADAPTATION	WHY IT MATTERS
Surfactant	
Branched bronchioles	
Extensive capillary beds	
Many small alveoli	

Why is it wrong to say that the round shape of alveoli gives the large surface area?

B3.1.5 How the lungs are ventilated

TEXTBOOK PP. 279–281

	INHALATION	EXHALATION
Diaphragm		
Abdominal muscles		
External intercostals		
Internal intercostals		
Ribcage		
Thorax volume		
Pressure in the thorax		
Air flow		

Why are different muscles needed to move the ribs in opposite directions?

B3.1.6 Measuring lung volumes

TEXTBOOK PP. 281–283

Sketch a spirometer trace with normal breaths, one deepest breath in and one fullest breath out. Label **tidal volume**, **inspiratory reserve**, **expiratory reserve** and **vital capacity**.

Define each term, and write the equation for vital capacity.

Record your own results from the practical.

TIDAL VOLUME / DM^3	VITAL CAPACITY / DM^3	VENTILATION RATE / BREATHS PER MINUTE

Before you finish

Tick the ones you can explain without notes. Circle anything to ask about.

- [ ] Why size makes gas exchange harder
- [ ] The four properties of an exchange surface
- [ ] Three ways gradients are maintained
- [ ] Four adaptations of the lungs
- [ ] What surfactant does and why
- [ ] Muscle actions in inhalation
- [ ] Muscle actions in exhalation
- [ ] Cause and effect in ventilation
- [ ] Reading a spirometer trace
- [ ] The vital capacity equation

What the Cell. Original material written for this course.